

Blockchain Governance in the Era of Artificial Intelligence

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Abstract— This paper describes Blockchain Governance (that is, Governance for Blockchains) in the era of Artificial Intelligence (AI). We first provide an overview of blockchain technologies as well as a summary of our previous work on Cyber Security, Data, Cloud, AI and Social Media Governance. We then discuss ten major points in Blockchain Governance. These include (i) decentralized authority, (ii) roles and responsibilities, (iii) policy management, (iv) handling risks, (v) security and privacy, (vi) integrity, (vii) dependability, (viii) scalability, (ix) integrating cloud and blockchain, and (x) AI for Blockchain and Blockchain for AI. Finally, we also discuss how Blockchain may be used for Governance (that is, Blockchain for Governance).

KEY WORDS: *Blockchain, Governance, Policy, Law, Regulation, Security, Privacy, Scalability, Risk Management, Artificial Intelligence*

I. INTRODUCTION

Blockchain has become a critical technology for many applications including in finance, banking, healthcare, supply chain and more. The tremendous benefits of a blockchain include providing security and integrity while ensuring that the applications are decentralized as well as consensus among all participants is obtained to make decisions. Essentially, there is no central control in a blockchain. All the participants in the blockchain network are considered equal and have equal authority. Because of decentralization and not central control, blockchains are not subject to many of the regulations in general. As a result, they have become an attractive platform to execute transactions for many applications especially in banking and finance to transfer funds as well as to carry out trades.

In spite of all the benefits of blockchains, not having any form of control means there is potential for collusion and corruption. For example, in a Blockchain, all the members are considered equal and they do not need permission to join the network. As a result, they are not verified for trustworthiness. The members not being verified means they could be malicious actors and collude to commit fraud. Furthermore, the members have to agree and arrive at a consensus to commit a transaction. This could be a lengthy process. When we live in a society, however much we want to be autonomous and not obey a central authority, there has to be some governance to avoid chaos and lawlessness.

In the same way, we need some form of control for blockchains. As a result, various types of blockchains have been developed including IBM's effort on Hyperledger and the emergence of enterprise blockchains such as Quorum that

have to conform to some control. One needs permission to be part of such blockchains (e.g., Hyperledger) and also some form of governance is needed to ensure that malicious actors are thwarted.

In this paper we will discuss how blockchains may be governed in such a way that they maintain the desirable properties (e.g., some form of decentralization and consensus) but at the same time ensure that malicious actors do not join the chain. That is, we need governance for blockchains. This paper examines various aspects of governance that we have examined for technologies such as cyber security, data, cloud and AI and determine whether the governance policies that we have discussed for these other technologies are applicable to blockchains. It is now considered that together with AI, Quantum, Cyber Security, Data Science, and the Cloud, Blockchain is a critical technology for many applications and therefore we need such a technology to be governed according to appropriate policies. We believe that AI will be central to the future development and governance of blockchains,

While much of the focus of this paper is on Governance for Blockchain (which we call Blockchain for Governance), Blockchains may also be used in the governance for various systems. That is, blockchains ensure decentralization and consensus systems including electronic voting systems and cloud-based operations. Therefore, we also briefly discuss Blockchain for Governance in this paper.

The organization of this chapter is as follows. Section 2 describes blockchain technologies. Our prior work on governance is summarized in Section 3. Governance of blockchains will be discussed in Section 4. We also briefly discuss blockchain for governance in Section 5. Summary and directions are provided in Chapter 5. For an overview of Blockchain technologies we refer to [4]. For some details on Blockchain Governance we refer to [15].

II BLOCKCHAIN TECHNOLOGIES

This section provides a brief overview of Blockchain technologies. Blockchain is essentially a decentralized ledger that executes transactions. The ledger consists of a collection of nodes called the mining nodes (also known as miners). Miners mine for blocks solving a complex mathematical problem. They have to show the proof of their work (called Proof of Work or PoW) and the miner that successfully mines a block gets a reward in term of bitcoins which is the currency usually traded in Blockchains. Users submit transactions that enter the ledger. Each transaction has to be placed in a block. If a block is not available then the miners start the mining process. Once a block is mined and the PoW is shown, all the nodes have to agree using one of the consensus protocols

(e.g., Byzantine consensus is one such protocol). After all the nodes (or a percentage depending on the protocol) agree, then the transaction is placed in the block. The block is replicated in every node in the system. After execution of the transaction, all the nodes have to arrive at a consensus before committing the transaction. A blockchain is essentially a chain of blocks. Once the block is mined and accepted, it points to the previous block in the chain. The first block is called the Genesis block. The name Blockchain essentially means a chain of blocks.

Bitcoin is the currency commonly used to trade on a blockchain. For example, if John wants to transfer 10 bitcoins to Mary, then 10 bit coins in John's wallet are transferred to Mary's wallet. Each entity in the blockchain (e.g., user, node, wallet) has a public and a private key and each wallet has an address. Any data that is encrypted with say John's private key can only be decrypted by his public key, and vice versa. The request to transfer the bit coins is executed as a transaction. The transaction has information such as the sender's address, the receiver's address (that is, the addresses of the wallets), and the amount to be transferred together with some other information. After the transaction is executed by a block and replicated at all the nodes, then it has to be committed via consensus (that is, using a consensus protocol) and subsequently recorded on the ledger.

We have given just the basic concepts in blockchains and focused mainly on the Bitcoin Blockchain. There are numerous topics in blockchain including the different types of blockchains such as the Ethereum blockchain, the different crypto currencies such as ether (used in Ethereum), and also the different blockchain platforms such as public (e.g., Ethereum) and private (e.g., IBM's Hyperledger) and various enterprise blockchains (e.g., Quorum). While there are many similarities between the Bitcoin Blockchain, the Ethereum Blockchain, a major difference is that Ethereum has the concept of Smart Contracts. These smart contracts are not transactions, but they are executed by transactions. They are essentially programs that are executed when certain conditions are met. A useful resource for Blockchains is [4].

III CYBER SECURITY, DATA, CLOUD, AND AI GOVERNANCE

We have conducted research on Cyber Security, Data, Cloud, and AI governance for close to a decade [27], [28], [29]. We started our work on Cyber Security governance by first examining the various aspects such as threat analysis, risk management, identifying the roles and responsibilities of the individuals, evaluation of the systems and subsequent certification and accreditation. In addition, the policies and procedures that must be enforced as well as determining the sensitivity levels of the data are also important aspects of cyber security governance [27]. We also discussed the role of the Chief Information Security Officer (CISO).

As part of our investigation of Cyber Security Governance, we also briefly examined Data Governance. Data is one of the most critical resources of an organization and it is imperative that it is governed properly. Data governance is about ensuring that the data is identified, collected, cleaned, and subsequently stored, managed, analyzed and shared. It is also important to track the data and that is determining the

provenance of the data. We need answers to questions such as who created the data? who accessed the data? We also argued that we need policies to govern functions such as data collection, data storage, data management and data sharing. In addition we also need policies for data deletion. Another aspect that we have investigated is the role of the Chief Data Officer (CDO). Should such an officer report to the Chief Information Officer (CIO)? Because of the emerging critical technologies, organizations need to examine the roles of each of these officers and come up with an organizational governance plan [27].

Next we examined Cloud Governance. Cloud is now providing much of the data and services for organizations and applications. Furthermore, the cloud is becoming central to AI to train the various AI models and host the applications. Clouds are managed by the Cloud service providers (CSP). We need answers to questions such as the following: What is the relationship between the cloud client and the service provider? When the data in the cloud has to be deleted, who should monitor this deletion process? Finally what is the role of the Chief Cloud Officer (CCO)? We also need to conduct threat analysis and determine the risks involved with respect to the cloud. More details are provided in [28].

Our investigation also included AI and Data Science Governance [29]. Many steps in data science governance is also part of data governance. AI Governance is now one of the most important topics due to the extensive use of AI in almost all applications. Do we need guardrails on the AI systems to ensure that they do not hallucinate and are not attacked? This is a topic of much debate. Some argue that if we develop responsible AI systems, then it will inhibit the full potential of AI due to the guardrails. If we do not, then how can we trust the results produced by AI? Also AI Governance procedures must ensure features such as Accountability, Transparency, Explainability, and Responsibility of the AI systems as mentioned by Google in their document on AI Governance as well as in [29]. Other topics we addressed include the role of the Chief AI Officer (CAIO). Due to the importance of AI, should this officer also handle data governance? What are the relationships between the CAIO, CISO, CDO, and CCO? Is it feasible for all these officers to report to the CEO? We have also addressed AI Governance as part of our investigation of AI for Cyber and Cyber for AI [31]. Our discussion in this work included AI for Governance (that is, how can AI support governance such as policy management) and Governance for AI (that is, AI Governance) which discussed how AI systems must be governed. Some details are also provided in [22].

We have also investigated governance for systems such as social media in the era of AI [32]. The main focus here is the role of the Chief Social Media Officer (CSMO) which may or may not coincide with the role of the Chief Marketing Officer (CMO). One of the main concerns is spreading disinformation in social media. There are groups that state that curtailing disinformation is violating free speech. There are those that state that incorrect information could be harmful to humans. These are legal matters that the lawyers have to work out. But as technologists, we need to develop processes so that the social media users have a positive experience and gain accurate knowledge.

While we have investigated aspects of governing Cyber Security, Data, Cloud, AI, and Social Media, we also need to include the investigations of governance for systems and technologies such as Blockchains and Quantum Computing. One aspect that is concerning is the fact that quantum computing can break the code of the various crypto algorithms. This means most quantum crypto algorithms must work before we can release the quantum machines. This is an important governance issue for quantum computing. We have recently begun an recent investigation on Blockchain Governance and we provide a summary in this paper. With the increasing use of blockchains for many applications it is crucial that we must set up policies and procedures for the proper governance of blockchains. However, this violates the goals of blockchains which are built on decentralization, consensus, and deregulation without the control of a single authority. This is the topic of Section 4. Various studies and guidelines have also been released by NIST (National Institute of Standards and Technology) on blockchains [19].

IV BLOCKCHAIN GOVERNANCE

In this section we will discuss several aspects of Blockchain governance (that is, how can blockchains be governed; and also called Governance for Blockchain). The topics to be discussed include how can blockchains be governed without a central authority as well as policy management and security for Blockchains. Below we provide ten major points in Blockchain Governance. Much of the discussion on scalability, security, and privacy are provided in [4]. The curriculum we have developed in Blockchains as well as our work are reported in [30], [1], [21], [12], and [10]. These works have influenced the concepts on Blockchain Governance in this section.

- (i) How can Blockchain be governed without a central authority and regulations?

The major properties of blockchains include decentralization and consensus. That is, there is no central authority to ensure the checks and balances. Control is decentralized and every participant in the blockchain has to agree in order for the transactions to commit. This is opposite to the way traditional banks operate where a central authority ensures that the transactions are consistent and one does not overspend. The question is, if there is no central authority in a blockchain, then who is responsible for the governance of the blockchain? Should everyone be involved in the governance? This is one solution. Another is to have a group of trusted participants in the blockchain carry out the functions of governance. A third is to ensure that governance is carried out via an off-chain mechanism (that is, outside of the blockchain). That is, while the nodes in the blockchain conduct decentralized governance, one needs some form of overall governance via an off-chain. This violates the operation of a blockchain but provides some form of governance which is important. An interesting article on centralizing decentralization is provided in [7].

- (ii) What are the roles and responsibilities so the various individuals?

Our previous work on cyber security, data, cloud, and AI governance has stressed the roles of CISO, CDO, CCO, and CAIO. The question is, should there be an officer for

blockchains or should a CISO be responsible for the governance of blockchains. On the one hand if the CISO is responsible for all aspects of security, then he/she can also handle the governance of blockchains. However, in large organizations, there may be multiple clouds, numerous systems and several blockchains to govern. Therefore it may be impossible for the CISO to manage all the systems. However, a CISO could have multiple officers reporting to him/her. And the Chief Blockchain Officer (CBO) could be one of them. The CBO is responsible to ensure all aspects of the blockchain operation including decentralization, consensus, security, integrity, dependability, and scalability [22]. In addition, he/she must also determine which operations are to be carried out by the main blockchain and which by the off-chains [34].

- (iii) How can be Policies for Blockchains be managed and executed?

Policies are at the heart of governance. Appropriate policies are needed for enforcing security, privacy, and availability. Policies have to be specified, managed and executed. Appropriate policies and policy specification languages are needed. The policy management techniques include ensuring the consistency of the policies. Policies are specified as rules that must be satisfied. Policies may be executed as smart contracts (e.g., in Ethereum) using a programming language such as Solidity. Access control policies may specify rules for accessing the data or executing the transactions. Note that in our work we have used the terms regulations, laws, and policies interchangeably. However, they are three distinct concepts and explained in [17]. The idea of reducing regulations for blockchains is a contradiction with respect to the properties of blockchains such as decentralization and consensus. However, such contradictions may be resolved by the CBO or someone at his/her level. Perhaps one could make exceptions and introduce more regulations during certain periods such as a financial crisis in the banking industry. That is, before giving out say mortgages, it has to be approved by a trustworthy central authority in addition to the nodes in the chain. Aspects of policy management in Blockchains are discussed in [9].

Closely related to policy/regulation/law enforcement is the cross border operation of blockchains. The nodes in a blockchain could be located across multiple states and countries. Each state/county has its own policies and regulations. For example, GDPR (General Data Protection Regulation) is an EU (European Union) Law that protects the personal privacy of individuals while HIPAA (Health Insurance Portability and Accountability Act) is a US Federal law that protects sensitive information about patients. Therefore, blockchains where the nodes are in say the US and EU have to satisfy the policies/regulations/laws of the corresponding countries. That is, the nodes in the chain in the US should ensure that HIPAA is not violated while those in the EU should ensure that GDPR is not violated. Enforcing different policies in the same chain is complex and more research is needed. Aspects of data sharing in clouds using blockchains are discussed in [20].

- (iv) How can Risks be Managed?

Managing risks is a crucial component of governance. Therefore, the first step is to identify the risks for a

blockchain. Risks include those for security, privacy, integrity, among others. Since risks are mainly associated with security, we will focus on risk management for security in blockchains. The first step is to identify all the security threats and vulnerabilities for blockchains. A risk management team must assess the threats and vulnerabilities. Second, a risk model has to be developed and analysis carried out. This is a hard task as one has to develop an appropriate risk model for blockchains. A simple approach is to determine the single loss expectancy of an attack; determine the number of times such an attack can occur say in a year and then calculate the annual loss expectancy [11]. Third, the risks are eliminated to an acceptable level and the status is discussed with the senior management. Fourth, the CBO (or someone at that level) must sign off on the use of the Blockchain accepting any residual risks. Finally, insurance may be purchased to handle the residual risks. Aspects of risk management for blockchains are discussed in [YANG2025].

(v) How can Security and Privacy be provided for Blockchains?

Security includes both confidentiality/privacy and providing solutions to cyber-attacks. Confidentiality and privacy are provided through the cryptographic techniques incorporated into the operation of blockchains. That is, each entity (e.g., node) in the blockchain has both a private key and a public key so that confidentiality and authenticity are provided. Cyber-attacks to the Blockchain is usually handled by the various solutions for detecting cyber-attacks including applying various AI/ML techniques (Artificial Intelligence/Machine Learning) (e.g., [8]). Solutions to security, confidentiality, and privacy can be carried out both in the main chain as well as in the off-chain to enhance performance. Various encryption techniques to ensure the confidentiality of the data could be carried out by off-chain. Similarly, the various AI techniques for detecting cyber-attacks to the blockchain could also be carried out via off-chains. More details on Blockchain Security and Privacy can be found in [33], [6]. While the above discussion focuses on ensuring security and privacy for blockchains, the use of blockchains to provide security and privacy for computing systems such as voting systems are discussed in [3].

(vi) How can Integrity be provided for Blockchains?

Integrity ensures the accuracy of the data and the transactions. The encryption of the data together with the public and private key mechanisms utilized provide ways to ensure the accuracy and the authenticity of the data and the transactions. In addition, the consensus and decentralization mechanisms also provides a way to ensure the integrity. This way if there is an error in the execution of a transactions, malicious or otherwise, the transaction will not be committed until all the nodes (or majority of the nodes specified by the protocols used) in the blockchain agree. Other integrity mechanisms such as tracking the provenance of the data could be handled by off-chains to determine when and where the error occurred. That is, tracking the provenance of the data provides the history of the data. Various aspects of Blockchain and data integrity are provided in [23].

(vii) How can a Blockchain be made Dependable?

Dependability includes secure, privacy, integrity as well as fault tolerance, and real-time processing. We have already addressed features such as security, privacy, and integrity which can be considered to be part of Blockchain Governance. In this subsection we will discuss other features such as fault tolerance, availability, and real-time processing. The consensus protocols will ensure some fault tolerance. That is, in order to commit a transaction, all nodes (or a majority of the nodes) must agree. Additional mechanisms of fault tolerance could be executed by off-chains to ensure scalability. Availability is about ensuring that the blockchain is available for executing the transactions. One way to ensure availability is to have a backup blockchain in case the primary blockchain is unavailable due to a crash. However, when the blockchain recovers, the data from the backup blockchain has to be recorded in the main chain. Real-time processing ensures that the timing constraints of the transactions are met. Numerous real-time processing algorithms have been discussed in the literature [5]. We need to examine them and incorporate appropriate algorithms into the execution of the transaction.

(viii) How can Blockchains be made Scalable?

While Blockchain-based transaction execution provides enhanced security, privacy, and integrity, the enhancements come at the expense of performance. This is because the encryption/decryption mechanisms central to blockchain as well as decentralization and consensus make the transaction execution process extremely slow. Therefore, scalability is a critical aspect for blockchain-based transaction execution. Scalability can be provided by optimizing the execution (e.g., execution of the smart contract) as well as utilizing technologies such high performance computing and the cloud to execute the transactions. Scalability can also be provided by executing certain functions via off-chains (e.g., policy processing and providing security solutions). Adaptable and Scalable blockchains are discussed in [33].

(ix) What is the role of the Cloud for Blockchain and Blockchain for Cloud?

As discussed under scalability, cloud is a useful tool to provide scalability for blockchains. That is, blockchains can be executed in a cloud. Clouds typically consist of multiple machines and therefore the various blockchain operations can be executed in parallel in a cloud. This means that the blockchain services are provided as Blockchain as a Service (BaaS) to the clients (i.e., the users). On the other hand various cloud computing functions such as cloud-based information sharing can benefit with the use of blockchains. For example, information sharing policies may be executed by smart contracts. Also, the various transactions executed in a cloud can benefit with blockchains providing features such as security, privacy and integrity. Integrating Blockchain and Cloud is discussed in [18].

(x) AI for Blockchain and Blockchain for AI?

One of the critical areas being explored is the integration of AI and Blockchains. That is, how can AI help with Blockchains and how can Blockchains help with AI? Several papers have been published recently on both aspects and is one of the critical areas of research at present. Articles on AI

for Blockchain including Machine Learning, LLMs, and Agentic AI can be found in [13], [14], [ROSS2024], and [8]. The applications of Blockchain for AI can be found in [24].

AI can help with Blockchain operations by analyzing the blockchain transaction execution data and determining whether there are any cyber-attacks. AI can also help with generating and optimizing smart contracts, and detecting vulnerability in smart contracts. AI could also help with ensuring the consistency of the various policies enforced on the blockchains as well as enhancing the governance of blockchains. That is, AI could be integrated into every aspect of the operation of the governance of blockchains. Blockchain can help AI by providing security, privacy and integrity. For example, for AI applications that can be decentralized (e.g., decentralize the execution of agents), blockchain technologies may be utilized to provide consensus among the agents. Lot of research needs to be carried out on AI for Blockchain and Blockchain for AI.

4. BLOCKCHAIN FOR GOVERNANCE

In the previous section we have discussed Governance for Blockchains (also known as Blockchain Governance. We address issues such security privacy, and policy management for blockchain. Another aspect of Blockchain and Governance is the use of Blockchain for Governance of various systema and applications. For example, systems for government applications must be highly secure, ensure confidentiality and the integrity of the data. Blockchain with its decentralization and consensus features could provide lot of support in the governance of such systems. In this section we briefly discuss some of the issues involved.

Blockchains are being explored for us in electronic voting systems. Electronic systems have many vulnerabilities and are subject to cyber-attacks. As a result, we cannot trust the results produced by these systems. However, when such systems are hosted on ledgers such as Blockchains, they can provide decentralized execution and consensus. Before arriving at a consents all the nodes participating in the ledger must agree. Also, to ensure confidentiality, authenticity and integrity, each entity (e.g., node, user, wallet) has a public key and a private key and encryption is used to provide such features, That is, Blockchain is used in the governance of the voting systems. Several papers have been published on this topic (see for example, [2]). It should also be noted that ensuring features like confidentiality/privacy of the data used in Blockchain based legers can be carried out by off chains and is discussed in [4].

One major issue with blockchain-based voting systems is performance. That is, blockchains are much slower than more centrally controlled systems for transaction execution. Voting systems have to execute millions of transactions and from this point of view Blockchain-based ledgers may not be suitable. Nevertheless, high performance computing systems including cloud-based systems as well as the quantum systems of the future could enhance the performance of transaction execution significantly. It should also be noted that it is critical that blockchain-based voting systems must handle cyber-attacks. AI/ML-based solutions are being explored for detecting such attacks.

Blockchains can be used to govern several other systems including information sharing systems among government-

based coalitions across multiple countries. Other applications of Blockchain for governance include applications in supply chain management and cyber physical systems. Some details are provided in [26], [16].

5. SUMMARY AND DIRECTIONS

This paper has provided an overview of Blockchain Governance (i.e., Governance for Blockchains). We have built on our previous work on cyber security governance, data governance, cloud governance, AI governance, and social media governance. It has included many topics such as (i) decentralized authority, (ii) roles and responsibilities, (iii) policy management, (iv) handling risks, (v) security and privacy, (vi) integrity, (vii) dependability, (xiii) scalability, (ix) integrating cloud and blockchain, and (x) AI for Blockchain and Blockchain for AI. For each topic, we have briefly discussed various issues involved as well as some of the solutions. We have also provided references for many of the topics discussed. We need to examine each of the topics and explore in detail the issues, challenges, and directions.

One of the critical areas in Blockchain governance is the use of AI for Blockchains and Blockchains for AI. Much of the focus in the literature has been on AI for Blockchains. That is, AI has to be incorporated into every aspect of the operations of Blockchains. We also need to conduct a more in-depth investigation on how Blockchains can help with the operation of AI systems such as Agentic AI systems to provide decentralization, security, privacy, and consensus among the multiple agents.

ACKNOWLEDGEMENTS: The research reported herein was supported in part by NIST grant number 60NANB24D143. Any opinions, findings, conclusions, and recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of NIST.

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